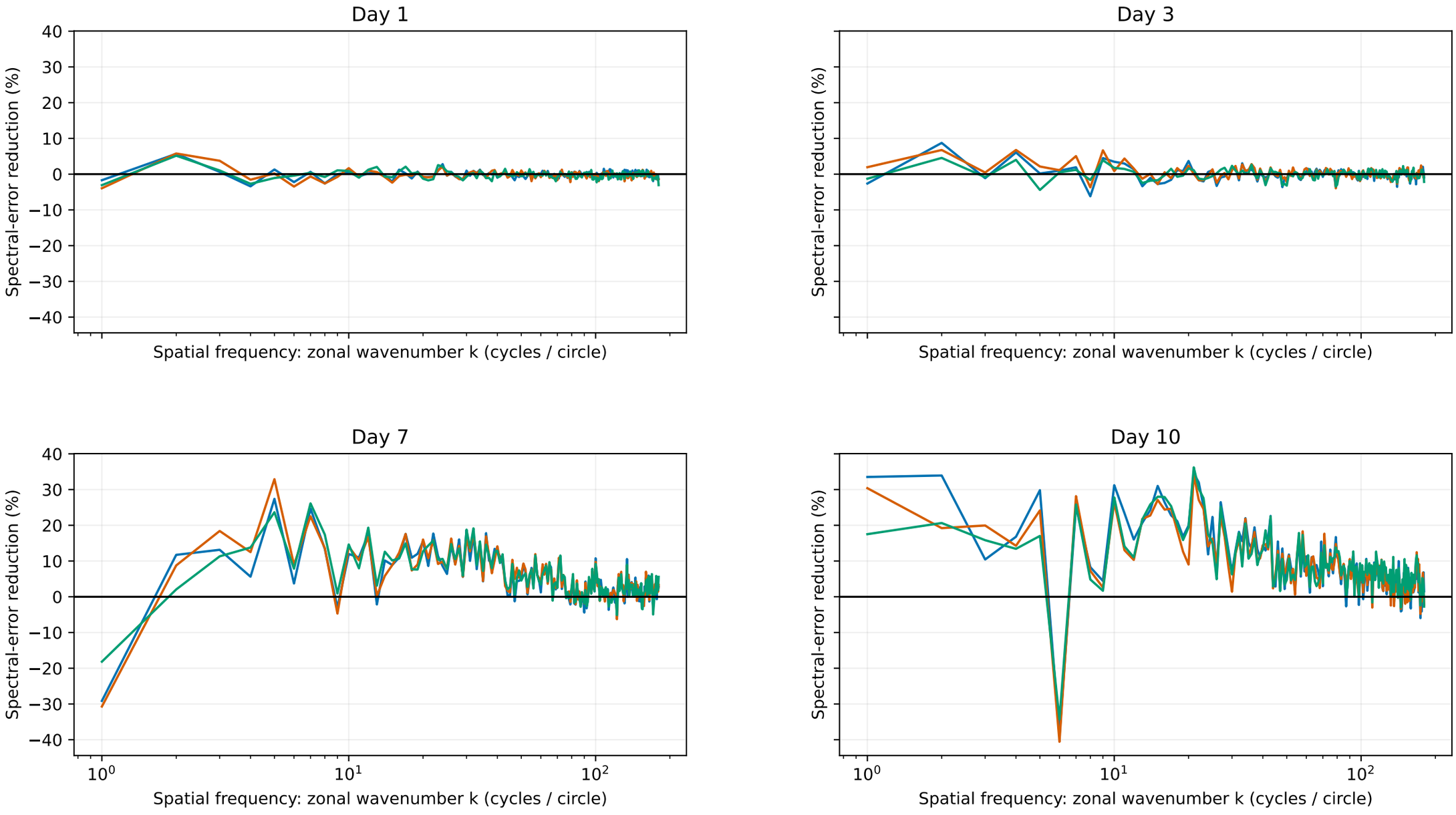
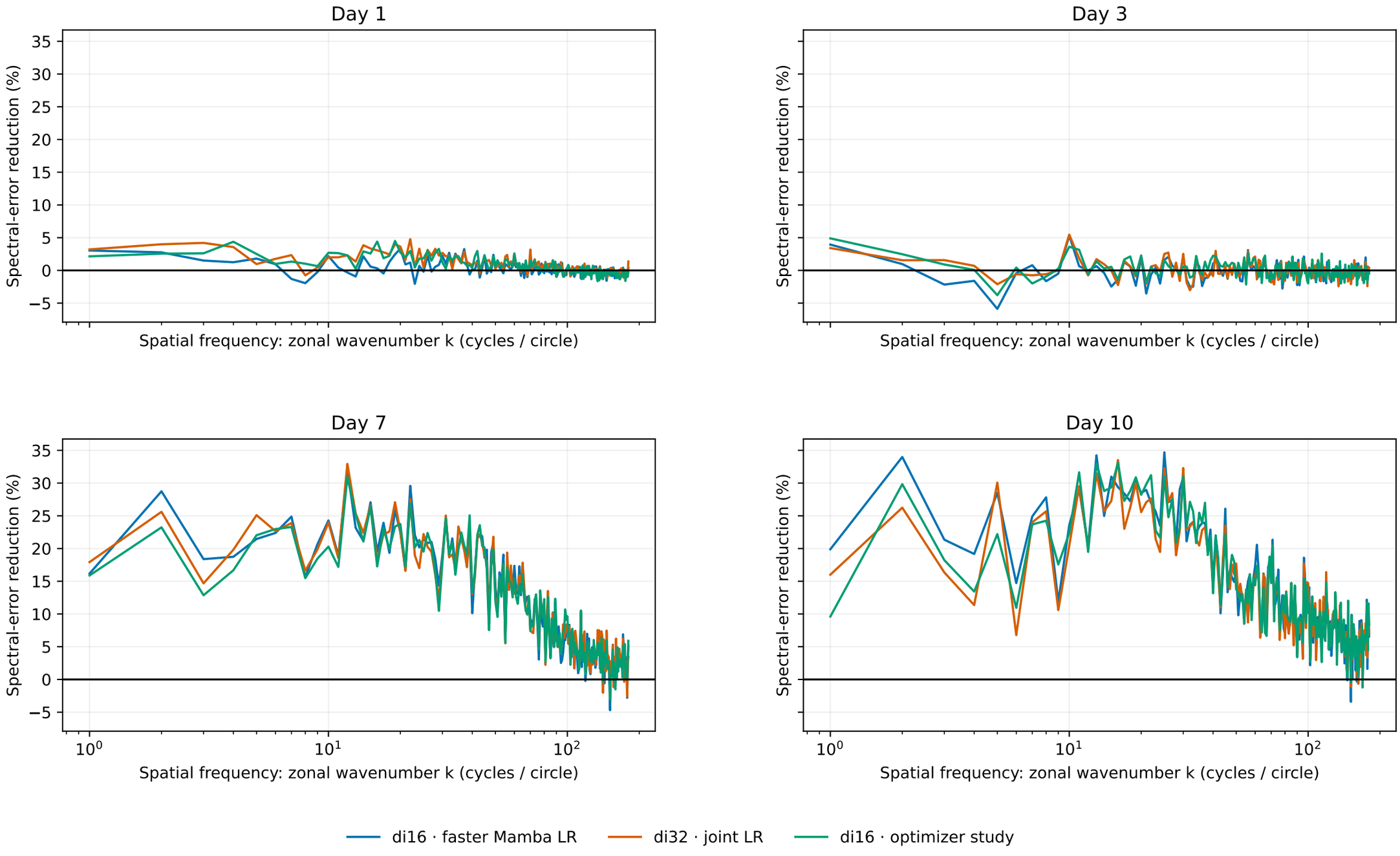
T2m: spectral-error reduction versus spatial frequency

Positive = better than original GC · Negative = worse · Zero = original GC



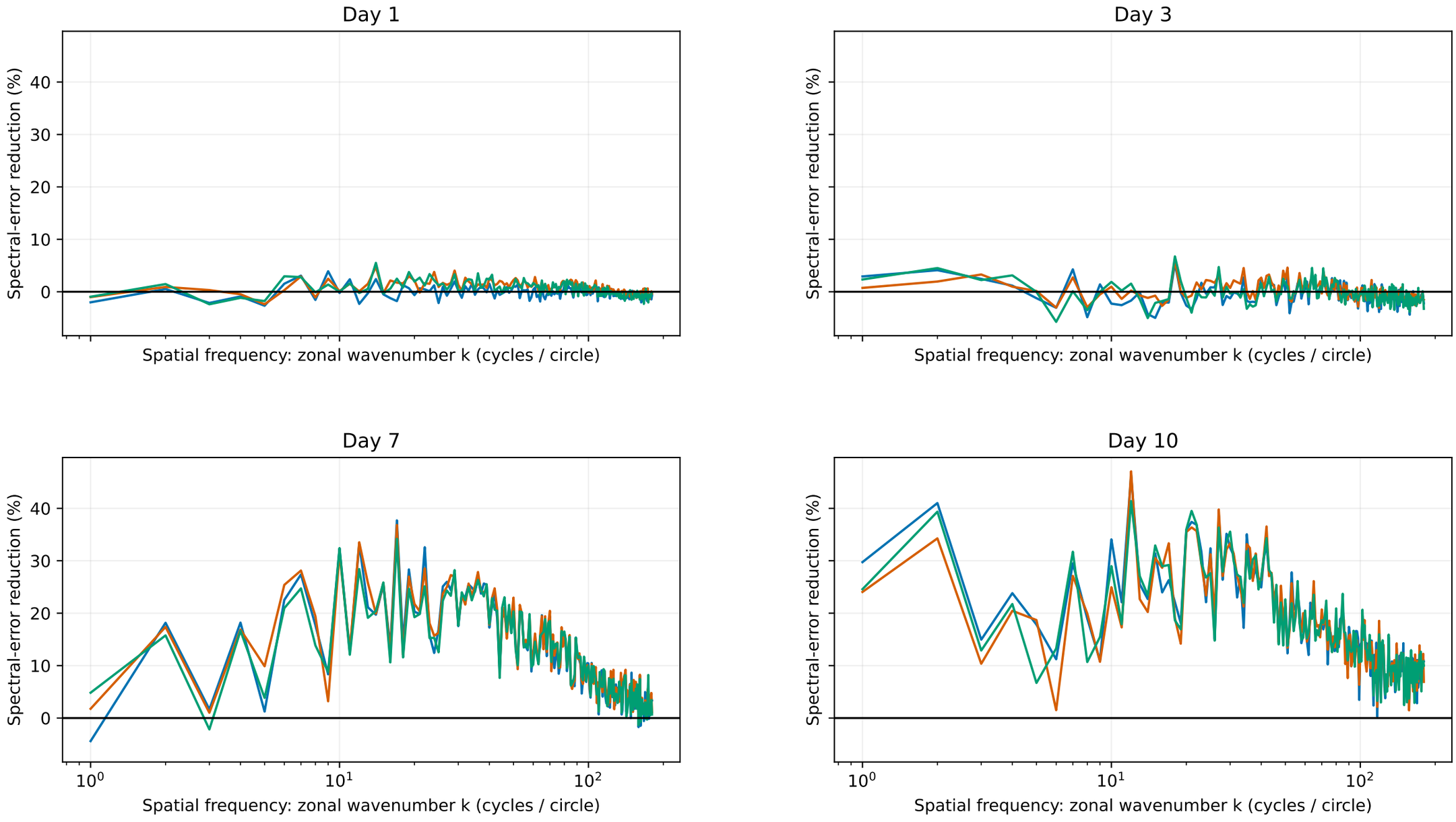
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


U10: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

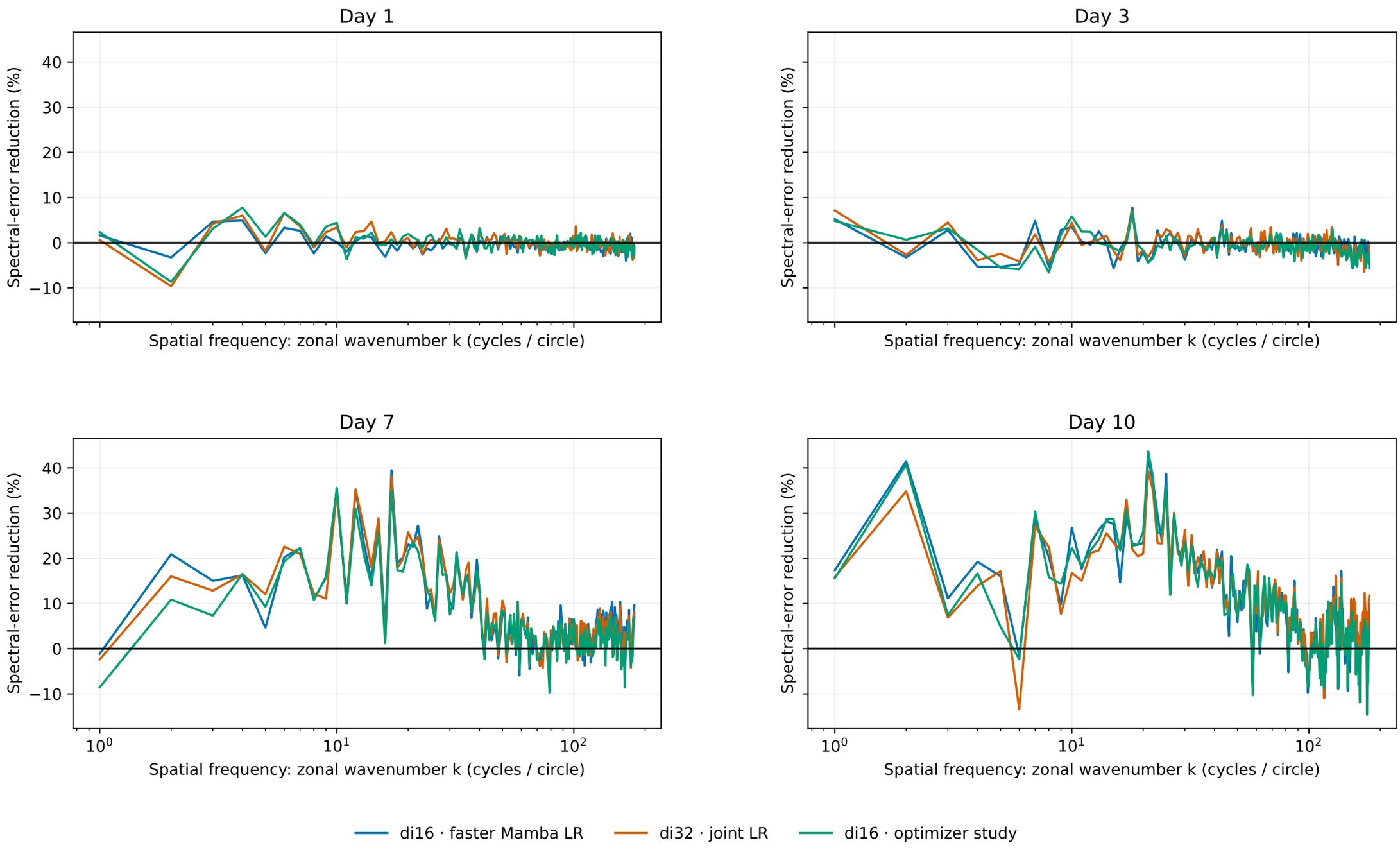
V10: spectral-error reduction versus spatial frequency

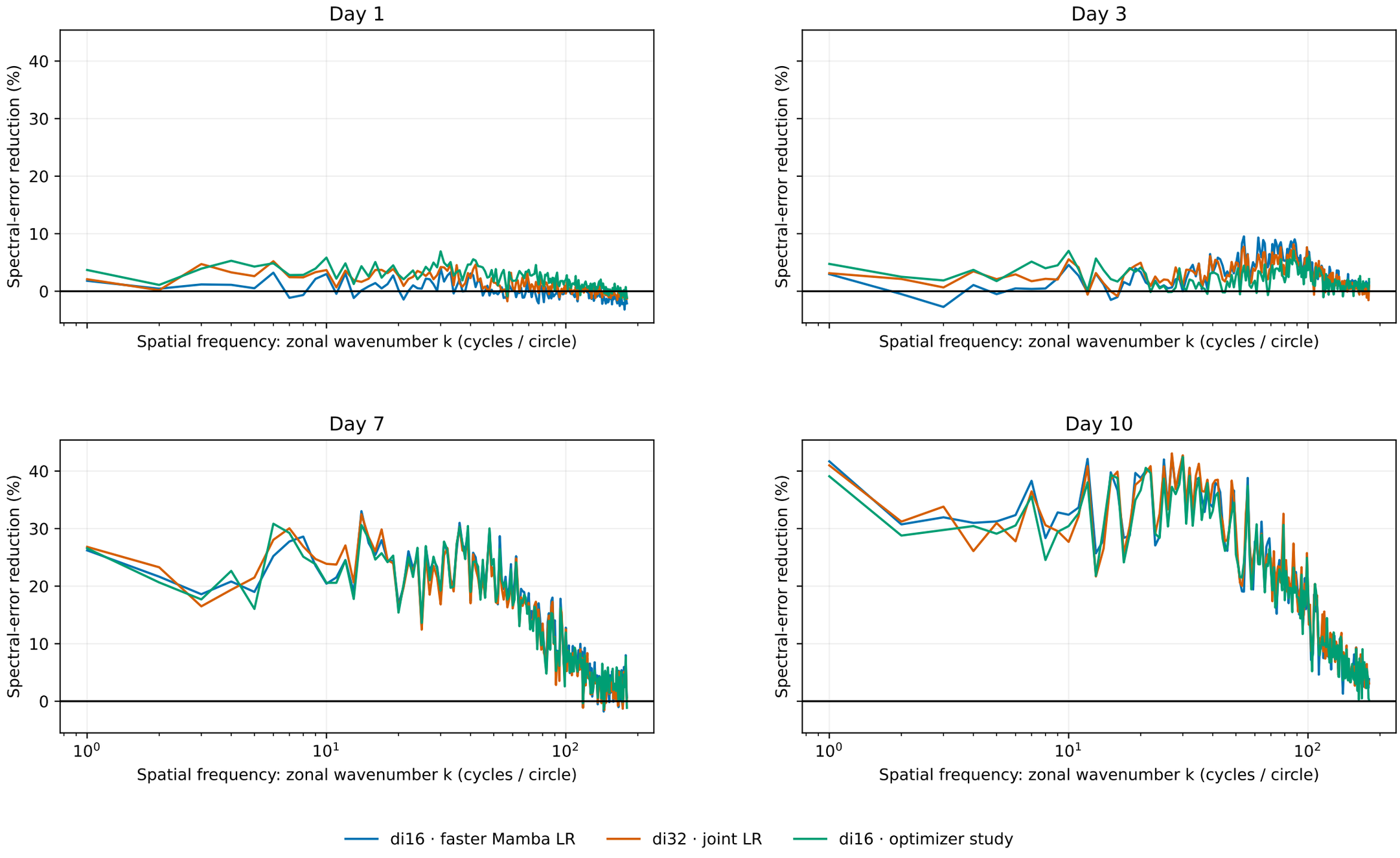
Positive = better than original GC · Negative = worse · Zero = original GC



1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

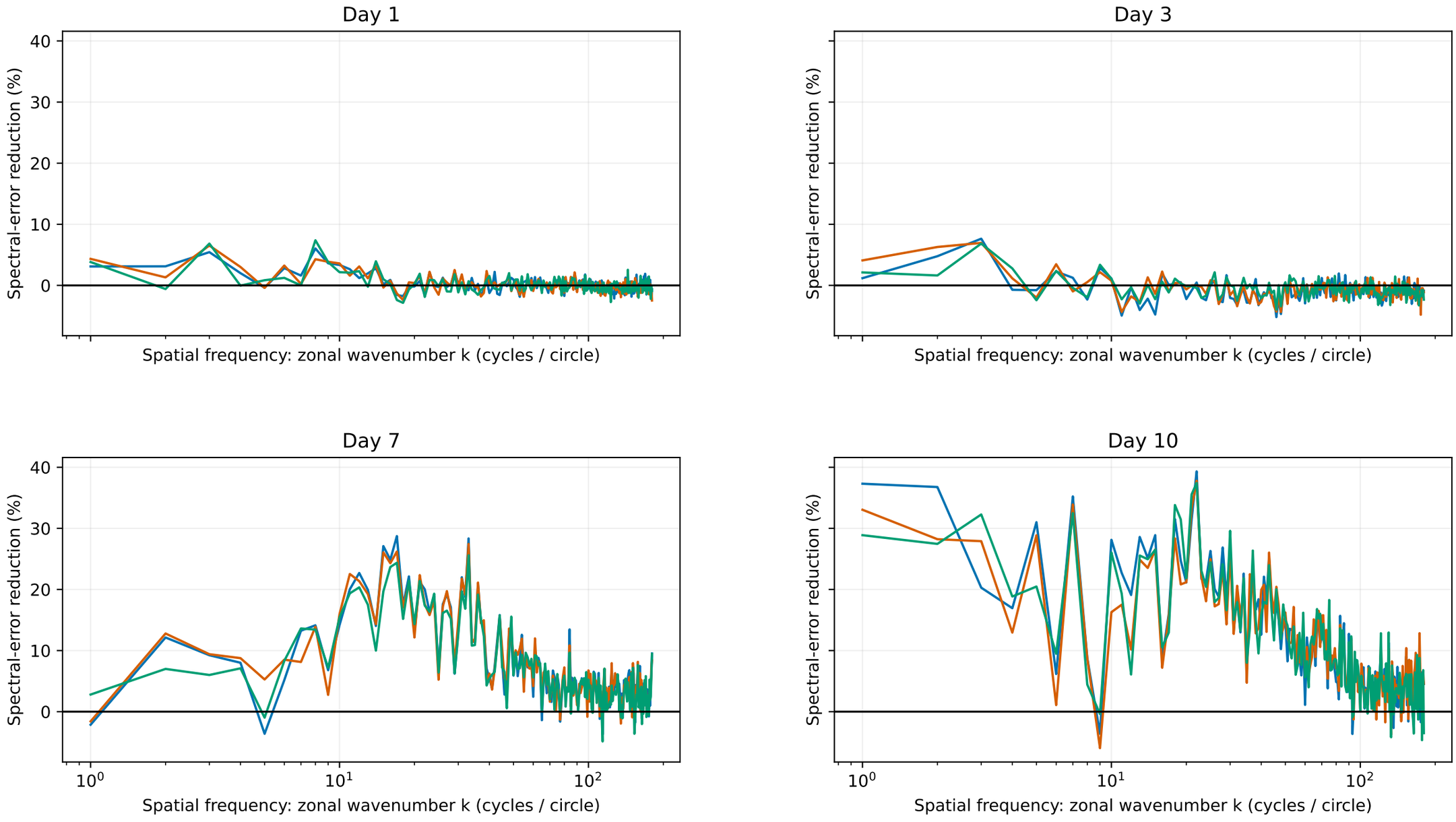
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


MSLP: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
40
30
20
10
0
−10
10^0
10^1
10^2
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


TP6: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.

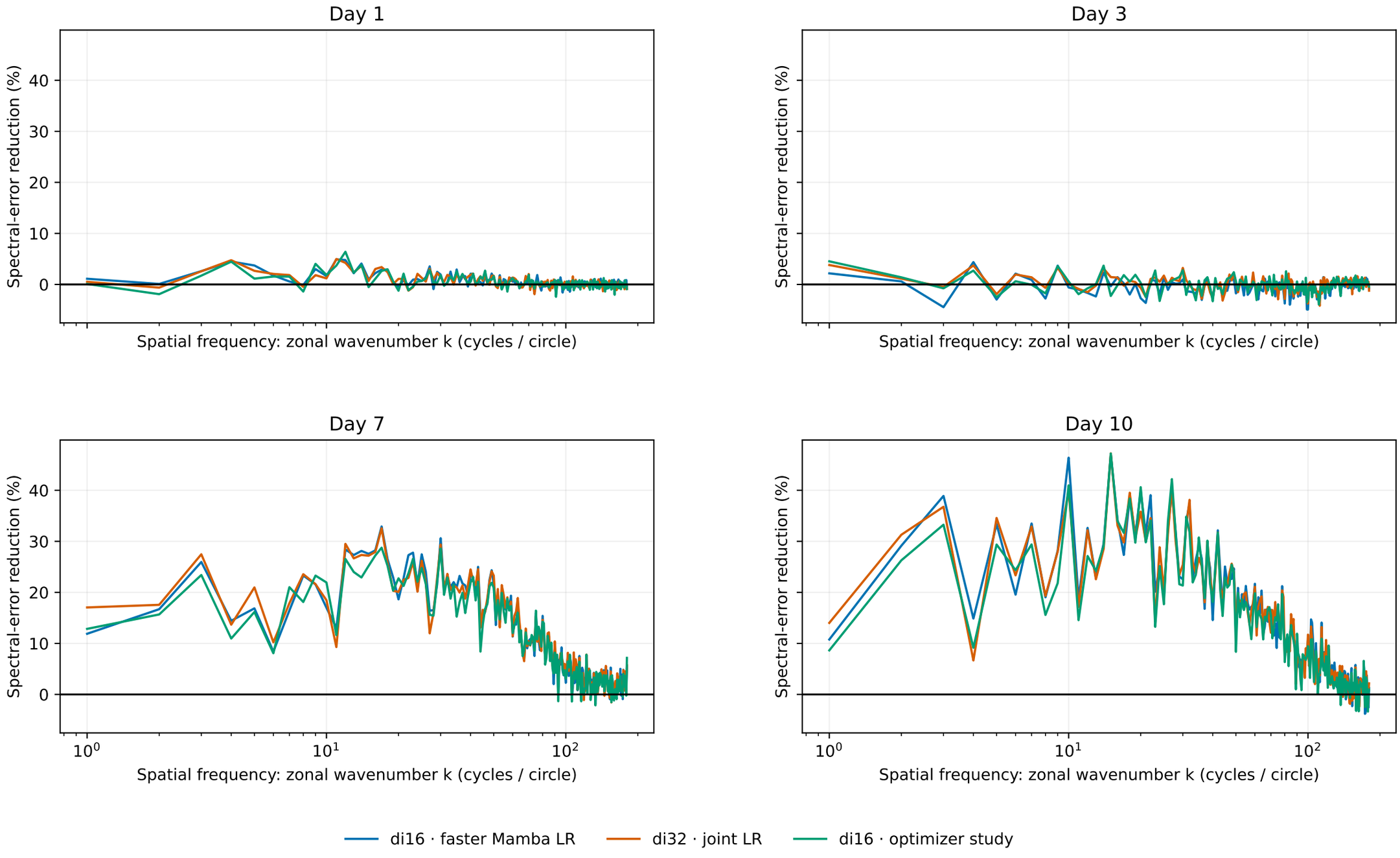
T850: spectral-error reduction versus spatial frequency

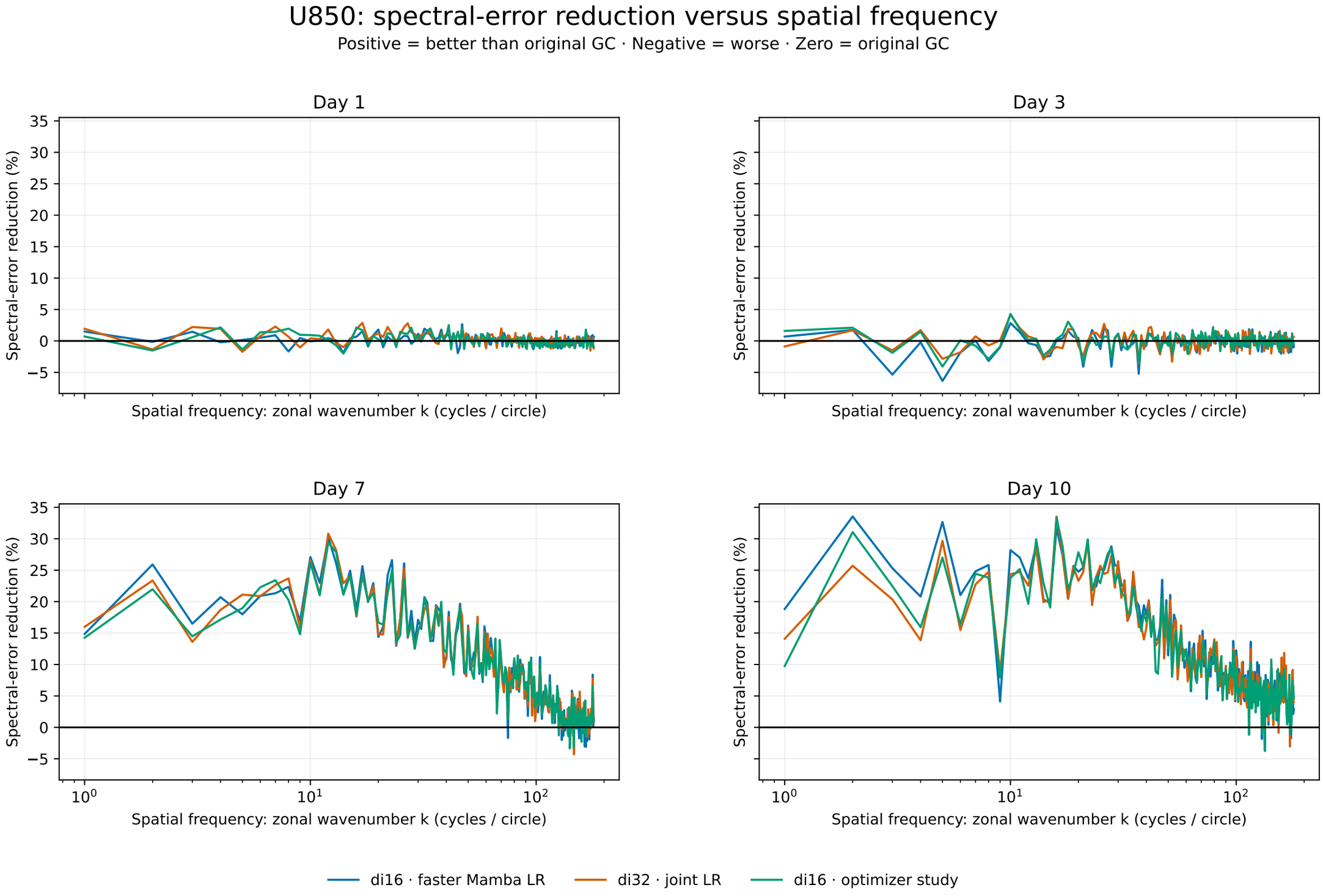
Positive = better than original GC · Negative = worse · Zero = original GC

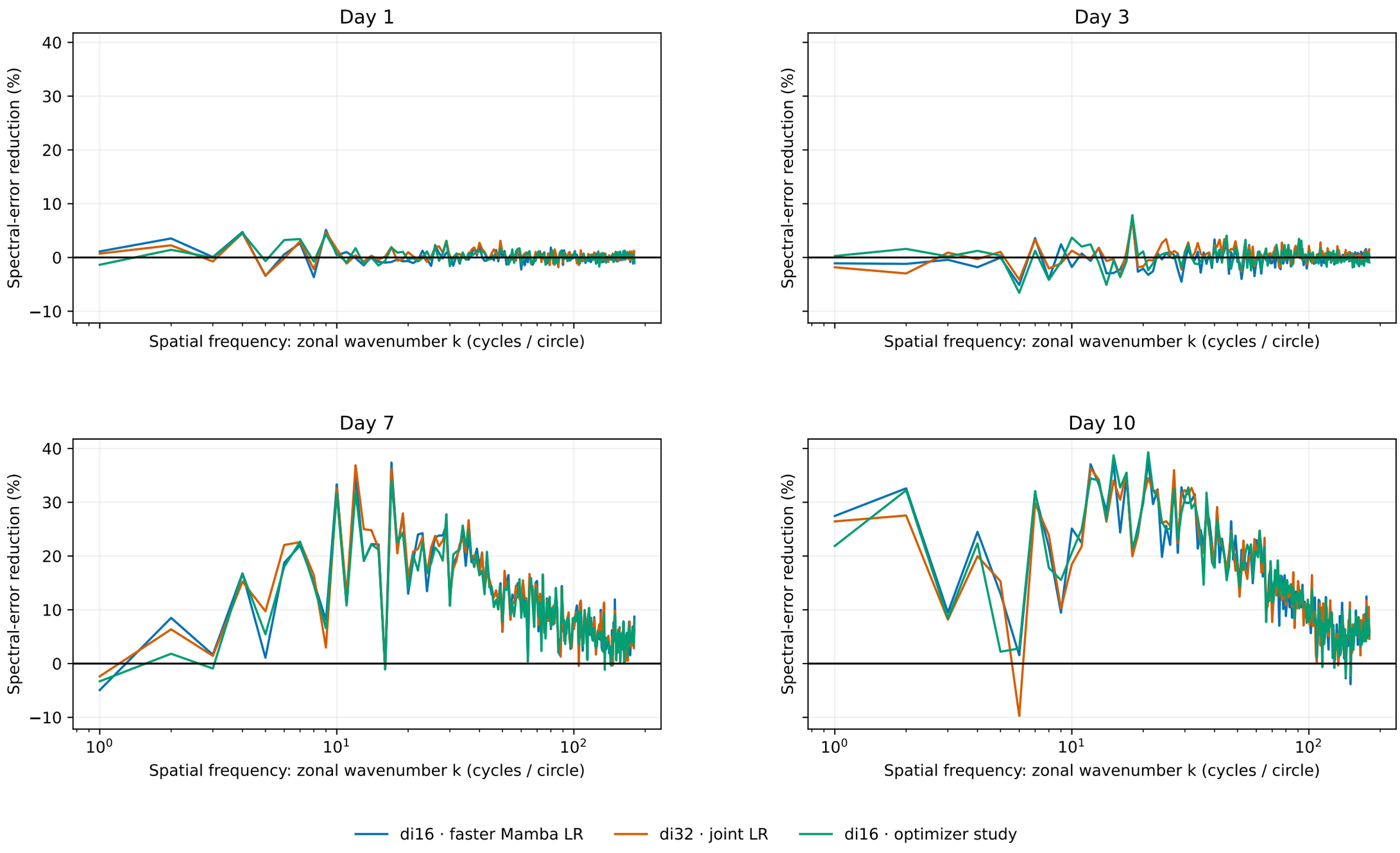


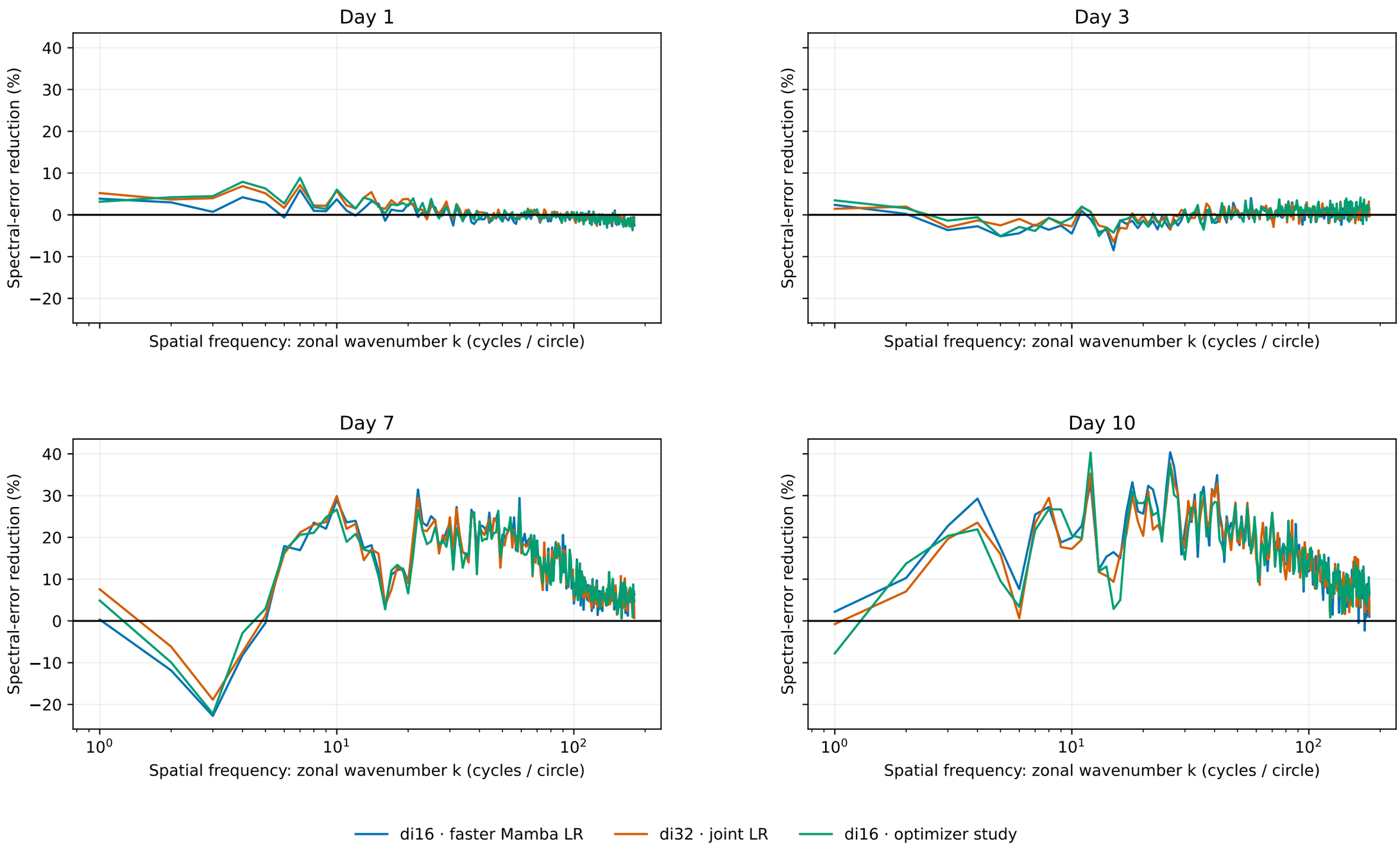
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.

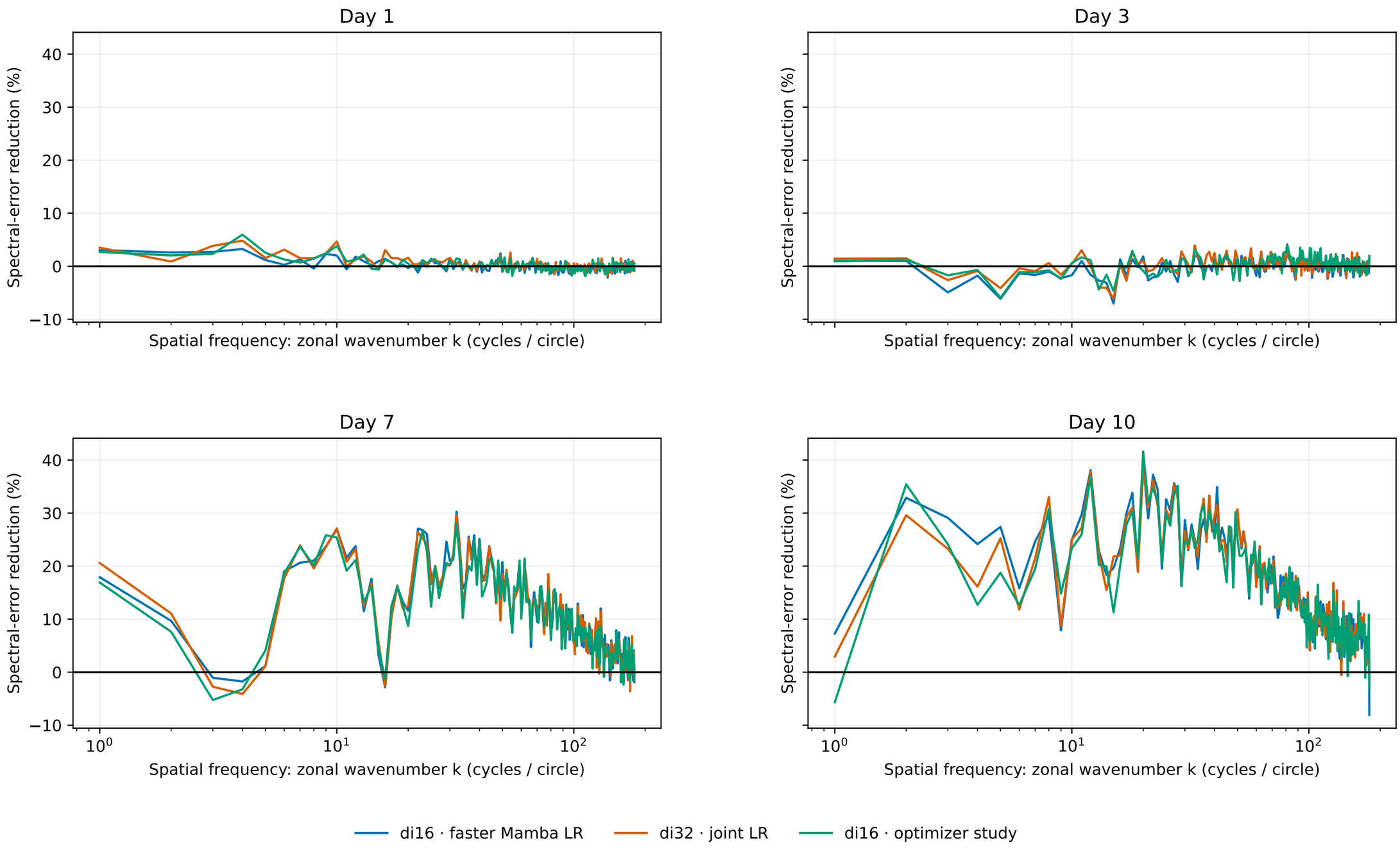
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


Q700: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


U850: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


V850: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


WS10: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.


WS850: spectral-error reduction versus spatial frequency
Positive = better than original GC · Negative = worse · Zero = original GC
Day 1
Day 3
Day 7
Day 10
Spectral-error reduction (%)
Spatial frequency: zonal wavenumber k (cycles / circle)
di16 · faster Mamba LR
di32 · joint LR
di16 · optimizer study
1° · Eight 2023 starts · 30–60° both hemispheres · Each model uses its own GC baseline.
Diagnostic: baseline repeatability unresolved; no uncertainty intervals. Spectral-error reduction is not exact GraphCast loss reduction.